

$$1) \underbrace{(-18) + 7}_{-11} - \underbrace{(-4) - 11}_{-5} = ?$$

$$-11 - (-5) = -16$$

$$2) \underbrace{[(-24) : 6]}_{-4} \cdot \underbrace{(-3)}_{-3} + \underbrace{15 : (-5)}_{-3} = ?$$

$$+12 - 3 = 9$$

$$3) \begin{aligned} (-3)^2 &= 9 \\ -3^2 &= -9 \\ (-2)^3 &= -8 \end{aligned}$$

$$4) \underbrace{4}_{4} - \underbrace{3 \cdot [2 - (1-5) \cdot 2]}_{-30} + \underbrace{18 : (-3)}_{-6} = ?$$

$$4 - 30 - 6 = -32$$

$$5) (-3)^2 - \underbrace{[(-24) : 3 + 2 \cdot (-5) - 6 \cdot (2-5)] \cdot 2}_0 = ?$$

$$9 - 0 \cdot 2 = 9$$

$$6) \underbrace{48 : 6 - 2 - 7 + 3}_0 = ?$$

$$16 - 7 + 3 = 12$$

$$7) x - \{1 - [2x - 3(x+1)]\} + 4(2-x) = ?$$

$$\begin{aligned} &\quad \quad \quad -x-3 \\ x - (x+4) + 8 - 8x &= 4 - 8x \end{aligned}$$

$$8) 27 \cdot 16 - 27 \cdot 6 = ?$$

$$27 \cdot \underbrace{(16-6)}_{10} = 270$$

$$9) a \text{ tek}, b \text{ çift ise } a^2 + 3b - ab = ?$$

$$\begin{aligned} a^2 &\text{ tek} & T + G + G &= \text{tek} \\ 3b &\text{ ve } ab && \text{ çift} \end{aligned}$$

$$10) 7 + 10 + 13 + \dots + 40 = ?$$

$$\frac{(40-7)+1}{3} = 12$$

$$12 \cdot \frac{(7+40)}{2} = 282$$

$$11) \text{Üç ardışık çift tam sayının toplamı 66, Sayılar kaçtır?}$$

$$\begin{aligned} x + (x+2) + (x+4) &= 66 \\ 3x + 6 &= 66 \\ 3x &= 60 \\ x &= 20 \end{aligned}$$

$$12) \text{Toplamı 20 olan sayının çarpımı en çok kaçtır?}$$

$$\begin{aligned} 10 + 10 &= 20 \\ 10 \cdot 10 &= 100 \end{aligned}$$

$$13) (243)_5 \text{ onluk sisteminde}$$

$$2 \cdot 5^2 + 4 \cdot 5^1 + 3 \cdot 5^0 = 73$$

14) 83 baslik taban?

$$\begin{array}{r} 83 \div 5 = 16 \text{ R } 3 \\ 16 \div 5 = 3 \text{ R } 1 \\ 3 \div 5 = 0 \text{ R } 3 \end{array} \quad (313)_5$$

15) $(132)_4 = 56$
 $4^2 + 3 \cdot 4 + 2 = 56$
 $12 + 13 = 56$
 $b = 6$

16) $\frac{(6! + 5!)}{4!} = \frac{(6 \cdot 5 \cdot 4! + 5 \cdot 4!)}{4!}$ 17) $-\frac{5}{6}, -\frac{3}{4}, \frac{1}{2}$ kesirlerin
büyüklüğü

$$6 \cdot 5 + 5 = 35$$

$$\frac{1}{2} > -\frac{3}{4} > -\frac{5}{6}$$

18) $\frac{5}{6} + \frac{4}{6} - \frac{2}{3} = ?$
 (5) (6) (10)

19) $\frac{7}{4} \cdot \frac{-23}{6} \cdot \frac{24}{11} = ?$

$$\frac{25}{30} + \frac{24}{30} - \frac{20}{30} = \frac{29}{30}$$

$$\frac{-161}{11}$$

20) $\left(2 \frac{1}{5}\right) : \left(1 \frac{1}{3}\right) = ?$

21) $\left(1 + \frac{1}{3}\right) - \left(\frac{1}{2} - \frac{3}{4}\right) = ?$

$$\frac{11}{5} : \frac{4}{3} = \frac{11}{5} \cdot \frac{3}{4} =$$

$$\frac{4}{3} - \left(-\frac{1}{4}\right) = \frac{19}{12}$$

$$\frac{33}{20}$$

22) $\left[\left(\frac{1}{2} + \frac{1}{3}\right) : \left(\frac{5}{6}\right)\right] - \left(\frac{3}{4}\right)^{-1} = ?$

23) $1 + 1 / [1 + 1 / ((1 + 1/2))] = ?$

$$1 - \frac{4}{3} = -\frac{1}{3}$$

$$\frac{8}{5}$$

24) $2,75 + 0,48 \cdot 2,5 - 12 \cdot 0,3$
 $2,75 + 1,20 - 4 = -0,05$

25) $\frac{0,02}{0,004} + \frac{0,03}{0,005} + \frac{2,04}{0,008}$

$$\frac{20}{4} + \frac{30}{5} + \frac{4}{8} = 5 + 6 + 0,5 = 11,5$$

26) 91 ve 97 sayılarının hangisi asal?
 $91 = 7 \cdot 13$ asal değil
 97 asaldır

27) 360 asal çarpanları
 $360 = 2^3 \cdot 3^2 \cdot 5$

$$\begin{array}{r} 360 \div 2 \\ 180 \div 2 \\ 90 \div 2 \\ 45 \div 3 \\ 15 \div 3 \\ 5 \div 5 \end{array}$$

28) 7 56 farklı asal bölenleri $\times$ tapanları
 2, 3 ve 7 $2+3+7=12$

29) 360 p.b.s?
 $t.b = 7$
 $t.t = 1$
 $p.b.s = (3+1)(2+1)(1+1) = 24$
 $t.b = 24 = 48$
 $t.t = 0$

30) a doğal sayı $N = 2^a \cdot 3^2 \cdot 5$ 24 pozitif bölme
 varsa a kaçtır?
 $(a+1) \cdot 3 \cdot 2 = 24$
 $a+1 = 4$
 $a = 3$

31) 72'nin kaç $2^a \cdot 3^b$ bölme
 vardır? 9 tane

$$\begin{array}{r} 72 \\ 1, 72 \\ 2, 36 \\ 3, 24 \\ 4, 18 \end{array}$$

32) 144'in kaç pozitif bölme tam bölünür?
 $144 = 2^4 \cdot 3^2$
 $2 \rightarrow 0, 2, 4$ $3 \cdot 2 = 6$
 $3 \rightarrow 0, 2$ $1, 4, 9, 16, 36, 144$

33) (84, 126) ebob?
 $84 = 2^2 \cdot 3 \cdot 7$
 $126 = 2 \cdot 3^2 \cdot 7$
 $ebob = 2 \cdot 7 \cdot 3 = 42$

34) (72, 120, 168) ebob?
 $72 = 2^3 \cdot 3^2$ $ebob = 2^3 \cdot 3 = 24$
 $120 = 2^3 \cdot 3 \cdot 5$
 $168 = 2^3 \cdot 3 \cdot 7$

35) (84, 126) ebob
 $84 = 2^2 \cdot 3 \cdot 7$
 $126 = 2 \cdot 3^2 \cdot 7$
 $ebob = 2^2 \cdot 3^2 \cdot 7 = 252$

36) (18, 24, 30) ebob
 $18 = 2 \cdot 3^2$
 $24 = 2^3 \cdot 3$
 $30 = 2 \cdot 3 \cdot 5$
 $ebob = 2 \cdot 3 = 6$

37) 12 ve 18 kalın 100'den küçük
 12 ve 18 ebob = 36
 36, 72, 108

$$38) \text{ebob}(48, b) = 12$$

$$\text{ekok}(48, b) = 240$$

$$b = ?$$

$$48 : b = 12 : 240$$

$$b = \frac{240 \cdot 48}{12} = 60$$

$$40) \text{ebob}(18, 30) = 6$$

$$\text{ebob} = (54, 30) = ?$$

$$\text{ekok}(54, 30)$$

$$\text{ebob} = 3, b = 18$$

$$\text{ekok} = 3 \cdot 30 = 270$$

$$42) (48, 60) \text{ ebob kaçtır?}$$

$$\frac{48}{12} = 4 \quad \frac{60}{12} = 5$$

$$4 + 5 = 9$$

$$44) (36, 120) \text{ ebob kaçtır?}$$

$$\frac{36}{24} = 1.5 \quad \frac{120}{24} = 5$$

$$1.5 + 5 = 6.5$$

$$46) 8 \text{ ve } 12 \text{ 'in ortak bölenleri}$$

$$\text{5'ın } 10 \text{ 'dan büyük olan en az katı?}$$

$$N \geq 10$$

$$N - 5 = 8k$$

$$N - 5 = 12k$$

$$24k = 5 + N$$

$$N = 101$$

$$48) \text{ girtilen } 4, b \text{ ve } 9$$

$$\text{en az kaç prizma}$$

$$\text{ekok}(4, b) = 36$$

$$\text{prizma sayısı} = \frac{36 \cdot 36 \cdot 36}{4 \cdot 9} = 216$$

$$39) \text{ebob}(a, b) = 6 \quad a + b = 24$$

$$\text{ekok}(a, b) = 80$$

$$a = 6x \quad 80xy = 6 \cdot 180$$

$$b = 6y \quad xy = 30$$

$$x = 5$$

$$y = 6$$

$$x + y = 5 + 6 = 11$$

$$41) (84, 126) \text{ ebob ile kaç parçaya}$$

$$42 \text{ ebob}$$

$$\frac{84}{42} = 2 \quad \frac{126}{42} = 3$$

$$3 + 2 = 5 \text{ parça}$$

$$43) (60, 84) \text{ ebob kaçtır?}$$

$$(60 + 84) : 2 = 288$$

$$\text{ekok} = 12 \quad \frac{288}{12} = 24$$

$$45) (12, 18, 30) \text{ de çözümler}$$

$$\text{ilk 3'ün en büyük tekrisi}$$

$$\text{ekok}(12, 18, 30) = 2^2 \cdot 3^2 \cdot 5 = 180$$

$$9 + 2 = 1200$$

$$47) 6 \text{ cm} \times 8 \text{ cm} \text{ dikdörtgen, parçalar}$$

$$\text{kağıt kağıt 5'ın en az katı}$$

$$(6, 8) \text{ ekok} = 24$$

$$\frac{24 \cdot 24}{6 \cdot 8} = 12$$

$$49) \frac{252}{360} \text{ en sade hal}$$

$$(36) = \frac{7}{10}$$

$$50) \frac{5}{18} + \frac{7}{24} - \frac{1}{12} = \frac{20}{72} + \frac{21}{72} - \frac{6}{72} = \frac{35}{72}$$

$$51) 540 \text{ k } \rightarrow \text{farklı karelerin k?}$$

$$540 = 2^2 \cdot 3^2 \cdot 5$$

$$k = 3 \cdot 5 = 15$$

$$52) A = 2^3 \cdot 3^2$$

$$B = 2^2 \cdot 3 \cdot 5$$

$$\text{ekok}(A, B) = 2^3 \cdot 3^2 = 12$$

$$\text{ebob}(A, B) = 2^2 \cdot 3 = 12$$

$$\frac{360}{12} = 30$$

$$53) (x, y) \quad \frac{x}{y} = \frac{3}{5}$$

$$\text{ekok} = 8$$

$$\text{ekok} ? \quad \text{ekok}(3t, 5t) = 8$$

$$t = 8$$

$$\text{sayılar } (2t, 4t) =$$

$$\text{ekok} \left(\frac{2t+4t}{8} \right) = 120$$

$$* 5 - 2 \cdot \left[\frac{3 - (2-6) \cdot 2}{11} \right] + 18 \cdot (-3) = ?$$

$$5 - 22 - 6 = -23$$

$$* 18 : 0,12 - 0,25 \cdot 8 = ?$$

$$\frac{180}{12} - 2 = 15 - 2 = 13$$

$$* 2x - \{ 3 - [x - 2(x+4)] \}$$

$$3 - x + 2x + 8$$

$$2x + 3 + x + 8 = x + 11$$

$$* 420 \text{ asal kare farkları bölünür}$$

$$420 = 2^2 \cdot 3 \cdot 5 \cdot 7$$

$$4 \text{ farklı}$$

$$* 5 + 9 + 13 + \dots + 49 = ?$$

$$\frac{49 - 5}{4} + 1 = 12$$

$$* 180 \text{ p.b.s. ve t.b. toplam}$$

$$180 = 2^3 \cdot 3^2 \cdot 5$$

$$(2+1)(2+1)(1+1) = 18$$

$$12, \frac{54}{2} = 324$$

$$* a \text{ doğal sayı } N = 2^9 \cdot 3 \cdot 7$$

$$20 \text{ bölen verirse } a = ?$$

$$(a+1) \cdot 2 \cdot 2 = 20$$

$$a+1 = 5$$

$$a = 4$$

$$* (231)_4 = 2 \cdot 4^2 + 3 \cdot 4^1 + 1 \cdot 4^0 = 45$$

$$* \left[\left(\frac{3}{4} - \frac{1}{6} \right) : \left(\frac{7}{12} \right) \right] + \left(\frac{2}{3} \right)^{-1} = ?$$

$$\frac{7}{12} : \frac{7}{12} = 1 + \frac{3}{2} = \frac{5}{2}$$

$$* \text{ekok}(96, 144) = 48$$

$$\text{ekok}(96, 144) = 288$$

$$96 = 2^5 \cdot 3$$

$$144 = 2^4 \cdot 3^2$$

$$\begin{aligned} * \text{ebob}(60, 6) &= 15 \\ \text{ekok}(60, 6) &= 180 \\ b &= ? \quad b = 45 \end{aligned}$$

$$60, 6 = 15, 180$$

$$\begin{aligned} * (72, 120) & \text{ekok, parça} \\ \text{ebob}(72, 120) &= 24 \\ \frac{72}{24} &= 3 \quad \frac{120}{24} = 5 \end{aligned}$$

$$3 + 5 = 8 \text{ parça}$$

$$\begin{aligned} * (60, 72) & \text{on a 2 bas} \\ \text{ekok } 24 \quad \frac{240}{24} &= 10 \text{ bas} \end{aligned}$$

$$\begin{aligned} * (15, 20) & \text{ekok } 60 \\ \text{ekok}(15, 20) &= 60 \\ 10, 10 + 60 &= 11, 10 \end{aligned}$$

$$\begin{aligned} * (3, 4, 6) & \text{on a 2 bas, parça} \\ \text{ekok}(3, 4, 6) &= 12 \\ \frac{12, 12, 12}{3, 4, 6} &= 24 \text{ parça} \end{aligned}$$

$$\begin{aligned} * \text{ebob}(x, y) &= 6 \\ \text{ekok}(x, y) &= 72 \\ \text{toplami en az } x + y &= ? \end{aligned}$$

$$6x, 6y \quad xy = 12$$

$$\begin{aligned} \text{ebob}(x, y) &= 1 \\ \text{en küçük toplam } (3 + 4) &= 12 \end{aligned}$$